

MATHEMATICAL ANALYSIS 1
HOMEWORK 1

(1) Prove that there are infinitely many prime numbers (Euclid's Theorem).

(2) Prove the following lemma by contradiction:

Lemma 0.1. *Let $I \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ that satisfies the following condition: for any $x_1, x_2 \in I$ with $x_1 < x_2$, the entire closed interval $[x_1, x_2]$ is a subset of I . Then I is an interval, i.e. I could be an open, closed, or half-open-half-closed interval, and one or both of its endpoints can be infinite.*

(3) Prove that the solution of the equation $x^2 = 7$ is irrational.

(4) Prove that for any set X and for any two subsets $A, B \subset X$, we have $(A \cup B)^C = A^C \cap B^C$.

(5) **Quantifiers.**

(a) Write the negation of the statement

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}_+ \quad \text{such that} \quad \frac{1}{N} < \varepsilon,$$

moving the negation all the way inside (i.e. your answer should not contain the symbol $\neg$).

Which of the two statements (the original or its negation) is true? Explain.

(b) Determine whether each of the following statements is true or false, and explain the difference between them:

(i) $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad \text{such that} \quad x < y,$ (ii) $\exists y \in \mathbb{R} \quad \text{such that} \quad \forall x \in \mathbb{R}, \quad x < y.$

(6) Prove by induction that $2^n \geq n^2$ for all $n \in \mathbb{N}$ with $n \geq 4$. Is the inequality true for $n = 3$?

(7) Solve the inequality: $\sqrt{|x^2 - 4|} - x \geq 0$.

(8) Let $A \subseteq \mathbb{R}$ be nonempty and bounded from above, and suppose that $s_1, s_2 \in \mathbb{R}$ both satisfy the two properties in the definition of the supremum of A . Prove that $s_1 = s_2$. (This is the uniqueness part of Proposition 2.8 in the lecture notes.)

Hint: prove by contradiction.

(9) Let $P \subseteq \mathbb{R}$ be the set of prime numbers.

(a) Is P bounded from above? from below? what is its infimum? supremum? do either the supremum or infimum belong to P ?

(b) Let $Q = \{x \in \mathbb{R} \mid x^{-1} \in P\}$. Is Q bounded from above? from below? what is its infimum? supremum? do either the supremum or infimum belong to Q ?

(10) Consider the following subset of $\mathbb{R}$:

$$A = \left\{ x \in \mathbb{R} \mid 0 \leq x < 1 \quad \text{or} \quad x = \frac{2n-3}{n-1}, \quad n \in \mathbb{N} \setminus \{0, 1\} \right\}.$$

Is A bounded from above? from below? what is its infimum? supremum? do either the supremum or infimum belong to A ?